

Total No. of Questions : 8]

SEAT No. :

PD4698

[Total No. of Pages : 2

[6404]-204

B.E. (Mechanical)

DYNAMICS OF MACHINERY

(2019 Pattern) (Semester - VII) (402042)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1) a)** A 5 kg of mass is attached to lower end of spring, whose upper end is fixed and vibrates with a natural period of 0.45 Sec. Calculate value of K. The same spring having stiffness of K is cut into two halves, and two springs are used to support mass of 2.5 kg. Calculate natural period in sec for new system when two springs are used in parallel and series. **[10]**
- b)** Discuss the elements of a vibratory system and explain with neat diagram mathematical modeling of a bicycle. **[7]**

OR

- Q2) a)** Define the term damping factor and explain with neat sketches, underdamped, overdamped and critically damped systems. **[8]**
- b)** An under-damped shock absorber is to be designed for a motorcycle of mass 200 kg such that during a road bump, the damped period of vibration is limited to 2 seconds and amplitude of vibration should reduce to one-sixteenth in one cycle. Find the spring stiffness and the damping coefficient of shock absorber. **[9]**

- Q3) a)** Derive an expression for the amplitude of vibration for a system having rotating or reciprocating unbalance. **[8]**
- b)** A machine part of mass 4 kg vibrates in a viscous fluid. Determine the damping coefficient when a harmonic exciting force of 50 N results in resonant amplitude of 250 mm with a period of 0.4 second. If the excitation frequency is 2 Hz, find the percentage increase in the amplitude of forced vibration when damper is removed. **[10]**

OR

- Q4) a)** A radio set of 20 kg mass has to be isolated from a machine vibrating with an amplitude of 0.05 mm at a speed of 500 cpm. The set is mounted on four isolators, each having a spring scale of 31400 N/m and damping coefficient of 392 N-s/m. Determine: **[10]**
- i) Amplitude of vibration of radio set
 - ii) Dynamic load on each isolator due to vibration.
- b)** Derive an expression for deflection of vertical shaft with a single rotor without damping. **[8]**

P.T.O.

- Q5) a)** Explain the concept of torsionally equivalent shaft and derive equation for its equivalent length. [7]
- b)** Determine the natural frequency and corresponding mode shapes of given system shown in Fig. 5b when $K = 40 \text{ N/m}$ and $m = 10 \text{ kg}$. [10]

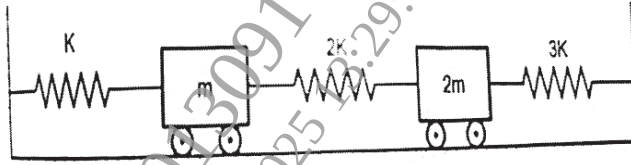


Fig. 5b

OR

- Q6) a)** Explain the torsional vibrations of geared system neglecting inertia of gears. [7]
- b)** For a spring-mass system shown in Fig. 6b, find Eigen values and Eigen vectors by matrix method. [10]

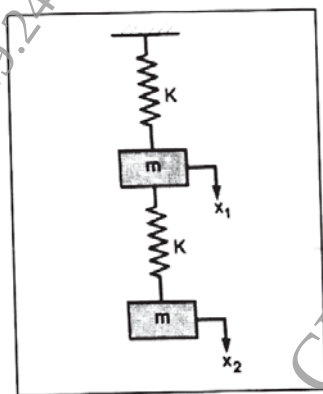


Fig. 6b

- Q7) a)** Explain Piezoelectric accelerometer with neat sketch. [6]
- b)** What is FFT Analyzer ? Explain how frequency spectrum can be used to detect vibration related problems in mechanical systems. [6]
- c)** Write short note on Seismic instruments. [6]

OR

- Q8) a)** What are different methods of vibration control? Explain any one in detail. [6]
- b)** Explain with neat sketch the working of sound level meter. [6]
- c)** Explain anechoic chamber and reverberant chamber. [6]

